Diffusion Process Augmentation in Semantic Segmentation

Yu-Chung Cheng, Yung-Lin Chiu
anthonycheng@vt.edu jimmychiu@vt.edu
Ismini Lourentzou, Instructor
lourentzou@vt.edu

Abstract

Diffusion model[4] is a generative model. We aim to provide a different aspect of using the diffusion model in machine learning. The goal is to deploy a diffusion model to add noise to the image and augment the data in semantic segmentation. This paper uses a lazy denoising function to retrieve the noise instead of deep learning to approximate the values. The experiment shows that it's almost as good as the detector without data augmentation in three classes of targets. However, in ten classes, the performances are pretty underfitting and the accuracy is low. The future work would be to try to use deep learning to get the noise in the denoising process. The code can be found at https://drive.google.com/drive/folders/10GyJ9tMIp4kc1Q0WptQFhfKfFoSQzz_P?usp=share_link.

1 Introduction

We try to provide a novel idea in data augmentation. The diffusion model is popular in generative AI (Artificial Intelligence). However, we believe it would be useful in other research areas. Therefore, we attempt to explore and implement the diffusion model into data augmentation.

1.1 Semantic Segmentation

Semantic segmentation is a pixel-level detector. It is helpful in autonomous vehicles and vision detection. However, labeling in pixels is time-consuming, so it's very important to reuse the label images with data augmentations to train the model. Thus, we decided to focus on noise-adding augmentation for semantic segmentation.

1.2 Diffusion Model

According to [3], the diffusion process is a sampling from a normal distribution (1) and deduces the image by steps, t, in (5). It has a closed-form equation in (6). Hence, we could use closed-form to calculate the noise in $O(n)$ time.

$$q(x_t|x_{t-1}) = N(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \tag{1}$$

$$q(x_{1:t}|x_0) = \prod p(x_t|x_{t-1}) \tag{2}$$

$$\alpha_t = 1 - \beta_t \tag{3}$$

$$\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i \tag{4}$$

$$q(x_t|x_{t-1}) = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon \tag{5}$$

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$$q(x_t|x_0) = N(\sqrt{\bar{\alpha}_t}x_0, \sqrt{1-\bar{\alpha}_t}I) \tag{6}$$

The most important work is the reverse process. It is necessary to use $\epsilon_\theta(x_t, t)$ to get the noise by the timestamp and apply (7) with it to get the recovered data. Normally, ϵ_θ would be the deep learning and trained by the noised data with a timestamp. However, we took a different approach and will introduce it in the methodology.

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)) + \sigma_t Z \tag{7}$$

1.3 Augmentation

There are multiple methods to implement data argumentation, such as flipping the image, resizing the image, and adding noise to the image. We only added Gaussian noise to extend our data set and compared the result of it to the result of the diffusion dataset. Our experiments showed that noise injection degraded the accuracy of semantic segmentation. The result is in line with expectation since noise injection will vague the boundary of the segmentation at the pixel level, and it would also result in underfitting.

2 Related Works

In [2], it used a diffusion process to add noise to the image and used a denoising process to recover the image. This system would extract the features from the image and get a better performance in medical image segmentation. Thus, it would be a good start for us to think about the application of the diffusion model.

3 Methodology

We use the diffusion model to add noise to the image and use the lazy-denoising process to recover the image from the noised one. In our original plan, we intended to use deep learning to learn the noise in each step, but it would take a lot of time, which would be even more than 1 GPU year for our data set. Therefore, we designed a simple way to guess the noise between each step. However, this process is not as reliable as deep learning to approximate the denoise function.

3.1 Diffusion Model

The most difficult part of implementing a diffusion model is the denoising process. It requires a training process in deep learning to predict the noise in each step. If the number of steps is 300, it would train the data for 300 times. Hence, we want to introduce a lazy-denoising process to handle this problem.

$$predicted_noise = q(x_t) - q(x_{t-1}) \tag{8}$$

3.2 Accuracy Calculation

The accuracy is calculated by the maximum arguments in each class by probability. We use this method to find the classes in each pixel and subtract them with the label image to get accuracy.

$$class = argmax_{classes} p_{classes}(x)$$

4 Experiments

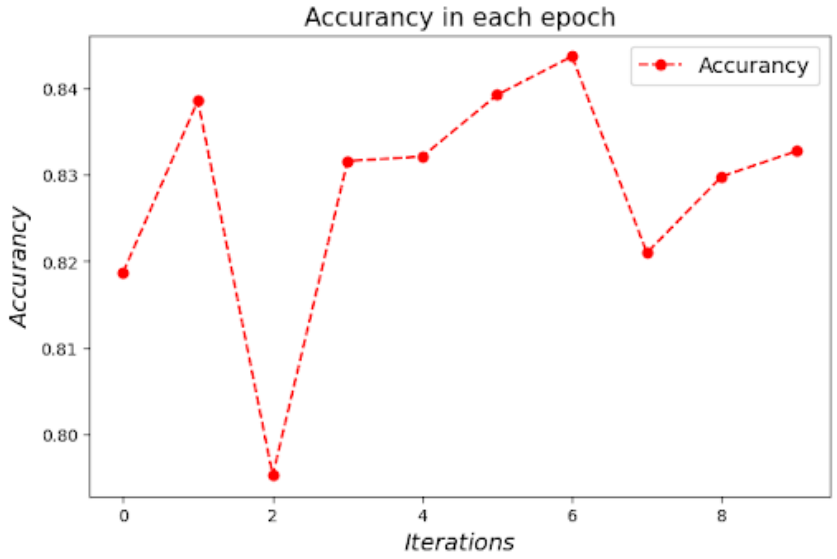
4.1 Data-set

We use [1] as our dataset. It's the most popular dataset in autonomous vehicle industries. Each image is high resolution in 2048×1024 pixels.

4.2 Results

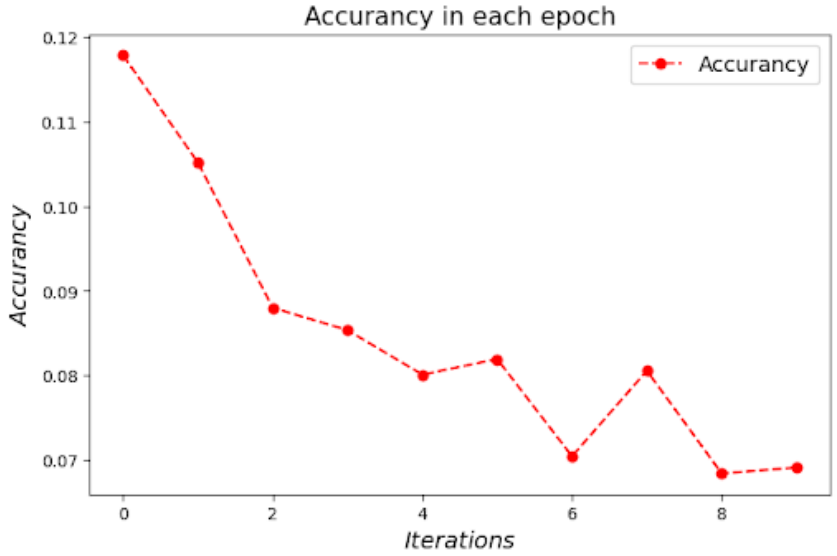
First, we do the experiments in three classes of targets.

Figure 1: Result of the default dataset



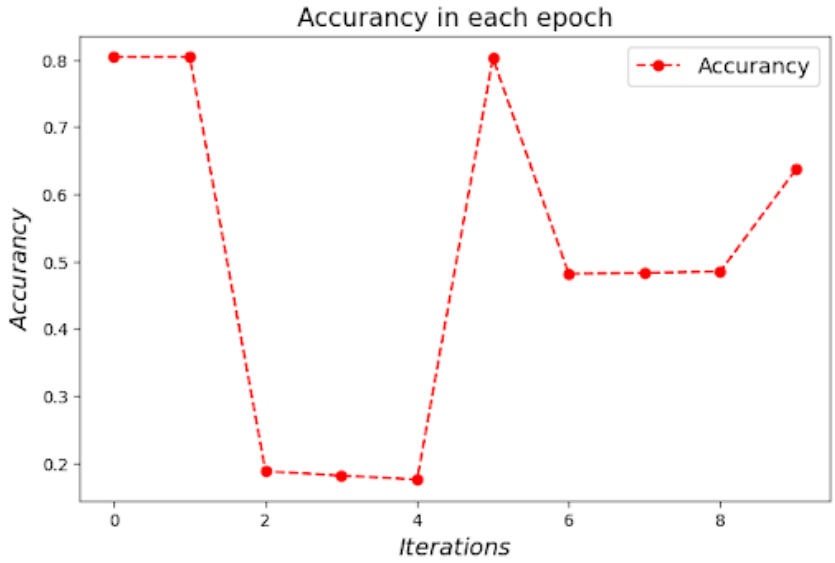
This is the result of the default dataset. Figure 1 shows the accuracy value of each epoch is generally great. The accuracy values fluctuate slightly between epochs, which is normal during the training process. This model is performing well.

Figure 2: Result of the dataset with Gaussian noise



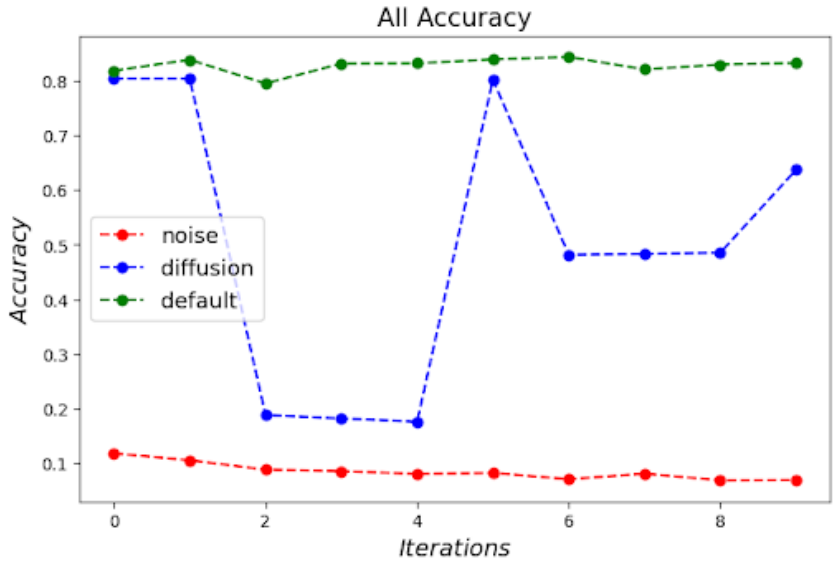
This is the result of the dataset with Gaussian noise. We only added Gaussian noise to the dataset. We have tried other kinds of noise, but the performances were generally bad. Thus, we decided to only pick the result of adding Gaussian noise to evaluate. Figure 2 shows that the result is bad, and it is clear that it is underfitting.

Figure 3: Result of the denoised dataset with Gaussian noise



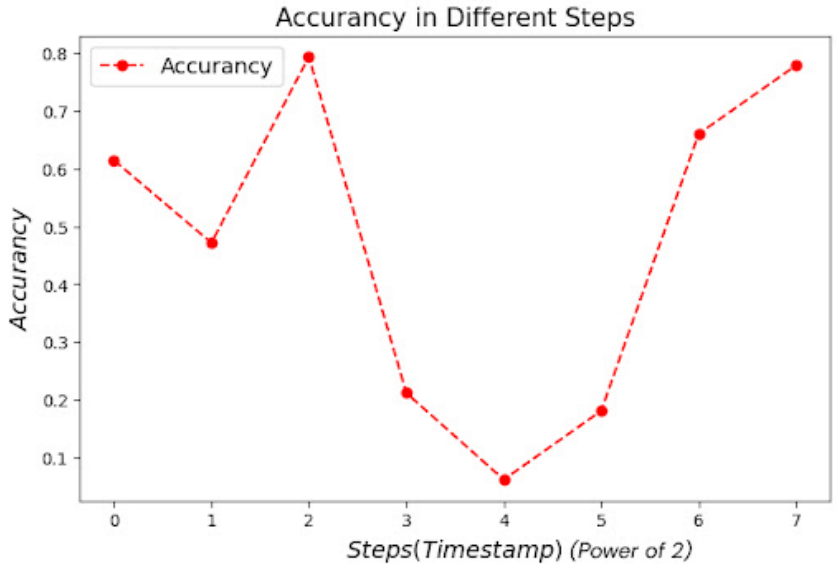
This is the result of the denoised dataset with Gaussian noise. Figure 3 shows that the accuracy value of each epoch is variable and unstable during the training process. The accuracy values fluctuate dramatically between epochs, but the result shows it was kind of better than the result of noise augmentation.

Figure 4: Comparison graph of different datasets



This is the comparison graph of different datasets. Figure 4 shows that the result of the default dataset is good, the result of the dataset with Gaussian noise is bad, and the result of the denoised dataset with Gaussian noise is unstable, while it is still a little better than the result of the dataset with Gaussian noise.

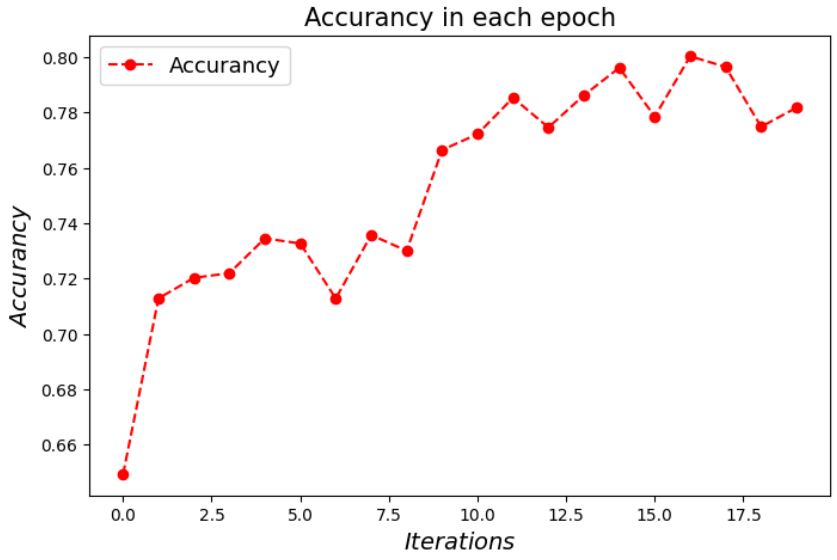
Figure 5: Result of the denoised (with different steps) dataset with Gaussian noise



This is the result of the denoised (with different steps) dataset with Gaussian noise. Figure 5 shows that the accuracy value of each epoch Is still variable and unstable during the training process. The accuracy values fluctuate dramatically between epochs, so our future work should be stable diffusion.

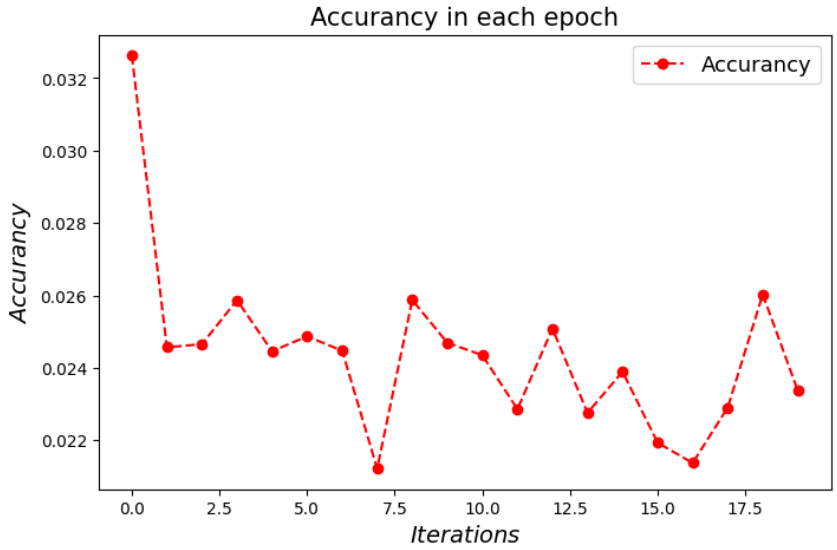
However, in the 10 classes, we got worse results on the diffusion model.

Figure 6: Result of the default dataset with ten classes



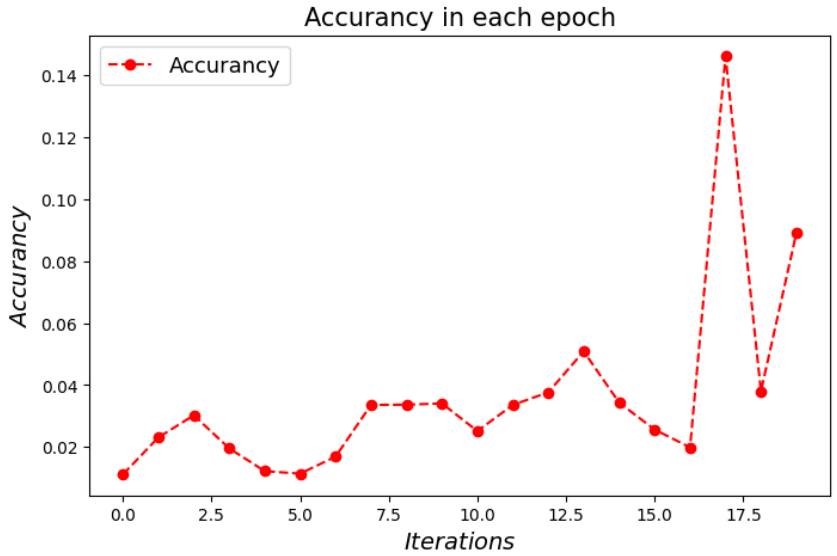
This is the result of the default dataset with ten classes. Figure 6 shows the accuracy value of each epoch is generally great, as we expected. The accuracy values gradually increase as more epochs, which is normal and good during the training process. This model is performing well.

Figure 7: Result of the dataset with Gaussian noise with ten classes



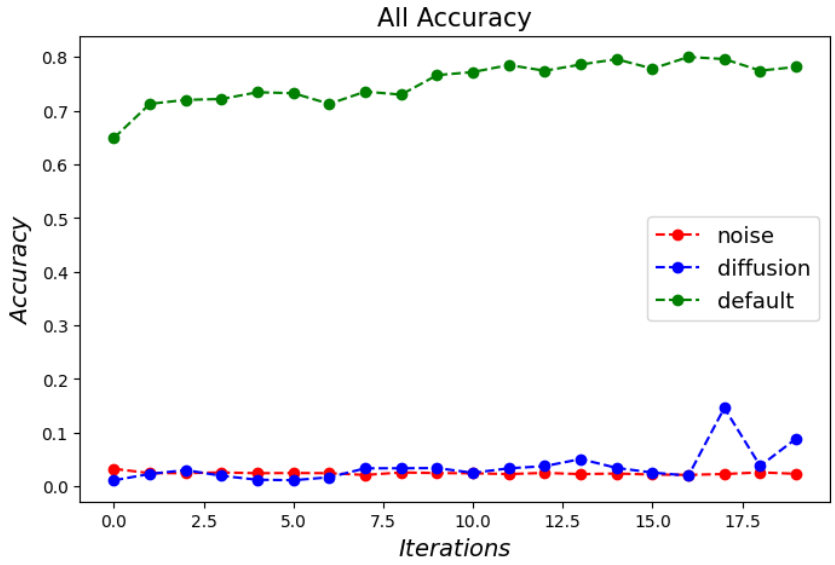
This is the result of the dataset with Gaussian noise with ten classes. We only added Gaussian noise to the dataset. We have tried other kinds of noise, but the performances were generally bad. Thus, we decided to only pick the result of adding Gaussian noise to evaluate. Figure 7 shows that the result is bad, and it is clear that it is underfitting.

Figure 8: Result of the denoised dataset with Gaussian noise with ten classes



This is the result of the denoised dataset with Gaussian noise with ten classes. Figure 8 shows that the result is bad, and it is even not better than the result of the dataset with Gaussian noise with ten classes.

Figure 9: Comparison graph of different datasets with ten classes



This is the comparison graph of different datasets with ten classes. Figure 9 shows that the result of the default dataset with ten classes is good, while the result of the dataset with Gaussian noise with ten classes and the result of the denoised dataset with Gaussian noise with ten classes are both bad.

Figure 10, Figure 11, and Figure 12 are visualization of different kinds of datasets with ten classes.

Figure 10: Visualization of the default dataset

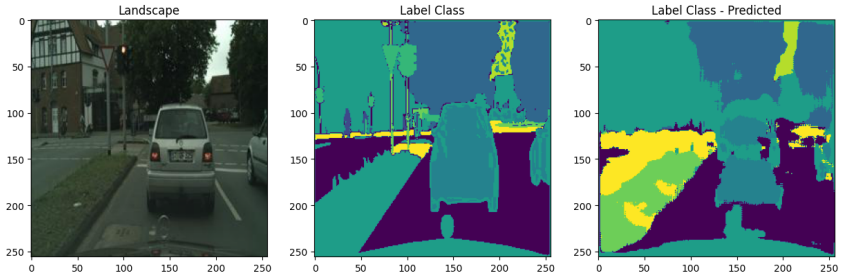


Figure 11: Visualization of the dataset with Gaussian noise

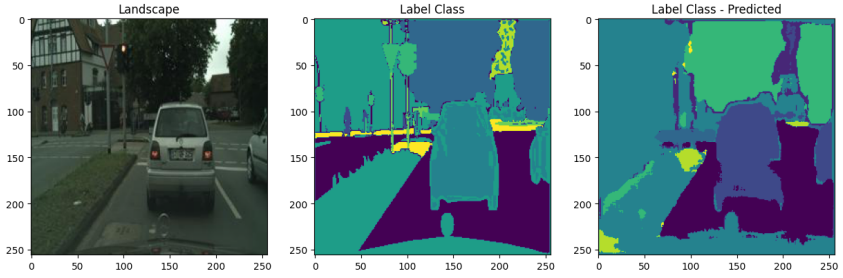
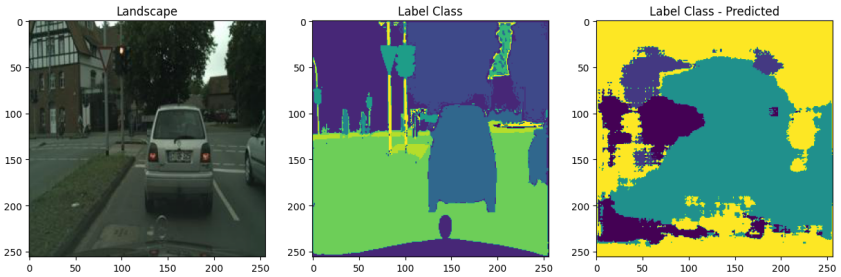


Figure 12: Visualization of the denoised dataset with Gaussian noise



5 Contribution

In our "Diffusion Process Augmentation in Semantic Segmentation" project, each team member made significant contributions to the success of the project. Yu-Chung played a crucial role in implementing the u-net architecture, diffusion model, and conducting experiments. Also, Yung-Lin made important contributions to the project by implementing adding noise, conducting experiments, and writing the experimental results. Our team's efforts and collaboration resulted in the successful outcome of our project.

References

[1] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The cityscapes dataset for semantic urban scene understanding. *CoRR*, abs/1604.01685, 2016.

[2] Aimon Rahman, Jeya Maria Jose Valanarasu, Ilker Hacihaliloglu, and Vishal M Patel. Ambiguous medical image segmentation using diffusion models, 2023.

[3] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *International Conference on Machine Learning*, pages 2256–2265. PMLR, 2015.

[4] Jascha Sohl-Dickstein, Eric A. Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. *CoRR*, abs/1503.03585, 2015.